

Project Blueprint: NifnityX

1. Project Overview

NifnityX is a real-time "Command Center" web interface for an AI-Enhanced Nifty Options Trading Bot.

The underlying trading engine (Python) analyzes market data using **Candlestick Patterns** and Machine Learning. NifnityX serves as the **control layer** for this engine. Its primary purpose is to visualize the market (Candlestick Charts), provide flexible execution workflows ("Safe Mode" or "Automated"), and allow strategy selection.

- **Primary Goal:** A low-latency dashboard to view live charts, monitor bot status, and authorize trades.
- **Target Audience:** Algorithmic Trader (Single User/Admin).

2. Technology Stack

The project uses the **MERN** stack optimized for performance and a specific dark aesthetic.

- **Frontend:**
 - **Framework:** React.js (Vite) - **Vanilla JavaScript (No TypeScript)**.
 - **Styling:** Tailwind CSS (v4) with **TweakCN / ShadCN True Dark Theme**.
 - **UI Architecture:** ShadCN Components.
 - **Charting:** lightweight-charts (TradingView Library) via a custom InfinityChart component.
- **Backend:**
 - **Runtime:** Node.js + Express.js.
 - **Auth:** JWT (HTTP-Only Cookies).
 - **Security:** bcrypt (hashing), crypto (AES-256 for API Key encryption).
- **Database:**
 - **DB:** MongoDB Atlas.
 - **ORM:** Mongoose.

3. User Roles & Permissions

The Trader (Admin)

- **Access:** Full access to all routes.
- **Keys:** Can add/update Broker API Keys (Angel One/Upstox).
- **Control:** Can toggle the Bot ON/OFF.
- **Approvals:** Can Approve or Reject trade signals generated by the bot.

4. Core Features (MVP)

4.1. Authentication

- **Login:** Email/Password access.
- **Security:** Route protection to ensure no unauthorized access.

4.2. Broker Connection ("Key Vault")

- **Global Mode:** The app supports **Paper Trading** and **Live Trading**.
- **Paper Trading Toggle:** A dedicated switch on this page: "Enable Paper Trading".
 - *If ON:* Hides/Disables the API Key inputs for trade execution (User might still need keys for Market Data if Angel One is the data source). The system will simulate order fills internally.
 - *If OFF:* Requires valid API Key, Secret Key, and TOTP/PIN.
- **Input:** Fields for API Key, Secret Key, and TOTP/PIN (specifically for Angel One/Upstox).
- **Validation:** "Test Connection" button to verify keys.

4.3. The Command Center (Dashboard)

The main workspace, divided into 3 specific functional zones:

A. Top Status Bar (The "Session HUD")

This bar is always visible so users never have to "count" trades manually.

- **Left Cluster (Market):**
 - **NIFTY 50:** Live Price (e.g., 24,500 +0.4%).
 - **INDIA VIX:** Live Volatility (e.g., 12.5).
- **Right Cluster (Session Health):**
 - **Trades:** Count (e.g., 3 / 5). *Visual: Simple text or mini progress bar.*
 - **Win Rate:** Percentage (e.g., 66%). *Visual: Green/Red text.*
 - **Net P&L:** Total (e.g., +₹1,500). *Visual: Large Badge.*
 - *Note: These stats toggle automatically between Paper/Live stats based on the global mode.*

B. Live Market Chart (Infinity Chart)

- **Visual:** A robust **Candlestick Chart** (using lightweight-charts) spanning the full width or center stage.
- **Data Source:** Fetches historical and live tick data from **Angel One API** (proxied via Node.js/Python).
- **Features:** OHLC Legend, Zoom/Reset controls, Live "Market Open" indicator.

C. "Safe Mode" Action Zone (The Dual Workflow)

This area behaves differently based on the **Strategy Tuner** setting (Auto vs. Manual).

- **Scenario 1: Manual Mode (Safe Mode)**

- **Trigger:** Bot detects a signal.
- **UI: Action Card** appears with "Signal Detected".
- **Content:**
 - **Header:** BUY NIFTY 25000 CE
 - **Context:** Strategy Name ("Aggressive") | AI Score ("78%").
 - **Smart Warning:** *Only appears if relevant (e.g., "Warning: Last allowed trade for today").*
- **Action:** User clicks [EXECUTE] or [REJECT].
- **Outcome:** Only *then* does the order go to the Broker (or Simulator).
- **Scenario 2: Fully Automated Mode**
 - **Trigger:** Bot detects a signal.
 - **UI: Notification Card** appears briefly: "Auto-Trade Executed".
 - **Content:** "Bought NIFTY 25000 CE @ 140".
 - **Action:** No user action required. The card just informs the user.
 - **Override:** A generic [STOP BOT] or [CLOSE ALL] button is always visible for emergency intervention.

4.4. Strategy Tuner (Redesigned)

A dedicated page to configure *how* the bot trades.

A. Execution Mode Selector (Crucial)

- **Toggle: "Manual Confirmation" vs "Fully Automated".**
 - *Manual:* Forces the Dashboard Action Card workflow.
 - *Automated:* Bot executes immediately upon signal.
- **Safety Lock:** Switching to "Automated" might require a second confirmation popup ("Are you sure?").

B. Strategy Profile Selector

- **Concept:** Select a "Profile" that maps to one of the 6 internal logic sets.
- **Options:** High Risk, Balanced, Conservative, Sniper.
- **Parameter Overrides:** Toggles for "Long Only", "Short Only".

4.5. Trade History (Hybrid)

- **Table:** Logs of all executed trades.
- **Smart Filters:**
 - **Source Filter:** [All] [Live] [Paper] (Crucial to separate real money from monopoly money).
 - **Execution Filter:** [All] [Auto] [Manual].
- **Columns:**
 - **Status:** (Win/Loss/Open).
 - **Mode:** (Paper/Live - represented by icon/badge).
 - **Type:** (Auto/Manual).
 - **Instrument:** (e.g., NIFTY 25000 CE).

- **Strategy:** (e.g., "High Risk").
- **Entry/Exit:** Prices.
- **P&L:** Profit/Loss.
- **ML Score:** AI Confidence.

4.6. Sidebar & Navigation (Redesigned)

A. Interaction Design

- **Collapsible:** The sidebar must support a collapsed state (Icons only) to maximize chart space on smaller screens.
- **Fixed:** It remains fixed on the left while content scrolls.

B. Top Section (Navigation)

- Logo (NifnityX)
- **Dashboard** (Home)
- **Analytics** (Deep Dive Charts)
- **Trade History** (Logs)
- **Strategy Tuner** (Config)
- **Broker Connection** (Keys)

C. Bottom Section (User Identity)

- **Profile Capsule:** A clean interactive area at the bottom.
 - **Visual:** User Avatar (Initials) + Email/Username.
 - **Action:** Clicks to open **Profile Page** (Summary stats).
- **Logout:** Distinct icon button next to the capsule.

4.7. Analytics & Visualization (New)

A. Profile Page (Summary View)

- **Goal:** Quick health check.
- **Visuals:**
 - **Equity Curve Sparkline:** Account growth over last 30 days.
 - **Win/Loss Donut:** Simple ratio.
 - **Key Stat Cards:** Total Profit, Max Drawdown, Risk/Reward Ratio.

B. Analytics Page (Deep Dive)

- **Goal:** Detailed performance study.
- **Data Source:** Aggregated from Trade History database.
- **Visuals:**
 - **Calendar Heatmap:** Day-wise P&L intensity (Green/Red grid).
 - **Daily Performance Bar Chart:** Magnitude of wins vs losses per day.
 - **Day-of-Week Analysis:** Which weekdays are most profitable?
 - **Strategy Comparison:** Bar chart comparing "High Risk" vs "Conservative" returns.
 - **Filtering:** Global toggle for **[Paper]** vs **[Live]** data.

5. System Architecture & Data Flow

Signal Flow (Live vs Paper)

1. **Python Engine:** Runs selected Strategy on Data.
2. **Signal Generation:**
 - *If Manual Mode:* Sends "Signal Proposal" event to Node.js -> Dashboard shows Action Card.
 - *If Auto Mode:* Sends "Order Execution" event immediately.
3. **Execution (Node.js Logic):**
 - *If Paper Trading:* Node.js calculates entry price based on live tick (simulated fill) -> Updates "Paper DB".
 - *If Live Trading:* Node.js calls Angel One API -> Updates "Live DB".

Chart Data Flow

1. **Source:** Angel One WebSocket/API.
2. **Proxy:** Node.js/Python subscribes to Nifty 50 tokens.
3. **Stream:** Emits OHLC packets via Socket.io to React.
4. **Render:** InfinityChart updates in real-time.

6. UI/UX Design System (The "True Dark" Palette)

This project strictly follows the **TweakCN / ShadCN** True Dark theme variables, enhanced with specific Modern SaaS requirements.

Key Design Principles

1. **True Dark Mode (Vercel-Inspired):** Use deep blacks (#000000, #090909) rather than standard grays to create a premium, high-contrast look suitable for long trading sessions.
2. **Modern SaaS Dashboard:** Use flat "Bento Grid" layouts or clearly defined Cards with subtle borders (#242424) to organize complex data. Avoid clutter.
3. **Icons:** Use **Lucide-React** for all iconography. They are clean, lightweight, and standard for modern React apps.
4. **Brand Identity:** Introduce a primary brand color which is modern, cool and suitable for our brand(make sure not too pick like any AI colors, pick like a professional designer and visionary founder of this system.) (e.g., Electric Indigo #6366f1 or Neon Blue #2962ff) for primary actions (Buttons, Active Links) to stand out against the black background.

Color Palette (CSS Variables): (Just for example not an actual one)

```
:root {  
  /* --- Default to Dark Mode colors --- */  
  --background: #000000;  
  --foreground: #ffffff;
```

```
--card: #090909;
--card-foreground: #ffffff;

--popover: #121212;
--popover-foreground: #ffffff;

--primary: #ffffff;
--primary-foreground: #000000;

--secondary: #222222;
--secondary-foreground: #ffffff;

--muted: #1d1d1d;
--muted-foreground: #a4a4a4;

--accent: #333333;
--accent-foreground: #ffffff;

--destructive: #ff5b5b;
--destructive-foreground: #000000;

--border: #242424;
--input: #333333;
--ring: #a4a4a4;

--sidebar: #121212;
--sidebar-foreground: #ffffff;
--sidebar-primary: #ffffff;
--sidebar-primary-foreground: #000000;
--sidebar-accent: #333333;
--sidebar-accent-foreground: #ffffff;
--sidebar-border: #333333;
--sidebar-ring: #a4a4a4;

/* Brand Specific Accents */
--brand-primary: #6366f1; /* Electric Indigo for Buttons/Links */
--brand-success: #22c55e; /* Trading Profit */
--brand-danger: #ef4444; /* Trading Loss */

--radius: 0.5rem;
}
```

Typography

- **UI Text:** Geist Sans or Inter (Clean, legible sans-serif).
- **Financial Data:** Geist Mono or JetBrains Mono (Monospaced alignment for Prices, P&L, and Logs).